

Class Notes II(Simplex Method)

Initial B.F.S., Simplex Table, and Computational Steps

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Introduction to the Simplex Method

A linear programming problem (L.P.P.) may have infinitely many feasible solutions. Therefore, checking every feasible solution to find the optimal solution is generally impractical.

The **Fundamental Theorem of Linear Programming** states that if an L.P.P. has an optimal feasible solution, then at least one **Basic Feasible Solution (B.F.S.)** is optimal.

Thus, instead of searching among all feasible solutions, we can search only among the basic feasible solutions.

The **Simplex Method**, also called the **Simplex Algorithm**, was developed by **George B. Dantzig in 1947**. It is an iterative procedure that starts with an initial B.F.S. and moves from one B.F.S. to another, improving the objective-function value until an optimal B.F.S. is obtained.

Why do we need the Simplex Method?

The graphical method is convenient when an L.P.P. has only two decision variables. For problems involving more than two variables, graphical visualization becomes difficult. The simplex method provides a systematic procedure for such problems.

The simplex method does **not** examine all feasible solutions. It moves systematically from one B.F.S. to another.

The simplex method can also indicate whether the problem has an unbounded solution.

Slack and Surplus Variables

Before applying the simplex method, inequalities are generally converted into equations by introducing additional non-negative variables.

Slack Variable: If a constraint has the form

$$a_1x_1 + \cdots + a_nx_n \leq b,$$

we add a non-negative variable:

$$a_1x_1 + \cdots + a_nx_n + s = b, \quad s \geq 0.$$

The variable s is called a **slack variable**.

$\leq$	$\longrightarrow$	$+$ slack variable
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Example:

$$\begin{aligned}
&\text{Maximize} && Z = 2x_1 + 3x_2 + 4x_3, \\
&\text{subject to} && x_1 + x_2 + x_3 \leq b_1, \\
&&& 2x_1 + 4x_2 - x_3 \leq b_2, \\
&&& x_1, x_2, x_3 \geq 0.
\end{aligned}$$

Introduce x_4, x_5 :

$$\begin{aligned}
x_1 + x_2 + x_3 + x_4 &= b_1, \\
2x_1 + 4x_2 - x_3 + x_5 &= b_2,
\end{aligned}$$

where $x_4, x_5 \geq 0$. Thus, x_4 and x_5 are slack variables.

Surplus Variable: If a constraint has the form

$$a_1x_1 + \cdots + a_nx_n \geq b,$$

we subtract a non-negative variable:

$$a_1x_1 + \cdots + a_nx_n - s = b, \quad s \geq 0.$$

The variable s is called a **surplus variable**.

$$\boxed{\geq \longrightarrow - \text{ surplus variable}}$$

Example

$$\begin{aligned}
&\text{Maximize} && Z = 2x_1 + 4x_2 + 4x_3, \\
&\text{subject to} && x_1 + x_2 + 2x_3 \leq b_1, \\
&&& 2x_1 + 4x_2 + 6x_3 \geq b_2, \\
&&& x_1 - 2x_2 + 4x_3 \geq b_3.
\end{aligned}$$

Introduce x_4 as a slack variable and x_5, x_6 as surplus variables:

$$\begin{aligned}
x_1 + x_2 + 2x_3 + x_4 &= b_1, \\
2x_1 + 4x_2 + 6x_3 - x_5 &= b_2, \\
x_1 - 2x_2 + 4x_3 - x_6 &= b_3.
\end{aligned}$$

Hence,

$$\boxed{x_4 \text{ is a slack variable}}$$

and

$$\boxed{x_5, x_6 \text{ are surplus variables.}}$$

Standard Matrix Form

After introducing slack or surplus variables, consider an L.P.P. with m constraints and N variables, where $N = m + n$. It can be written as $\boxed{\text{Maximize } Z = \mathbf{c}\mathbf{x}}$ subject to

$$\boxed{A\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq 0.}$$

Here,

$$A = [a_{ij}]_{m \times N},$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}, \quad \mathbf{c} = \boxed{\mathbf{c} = (c_1, c_2, \dots, c_n, 0, \dots, 0)}.$$

The zero coefficients normally correspond to the added slack or surplus variables.

Columns of A

Let the j -th column of A be denoted by $\boldsymbol{\alpha}_j$, $j = 1, \dots, N$. Then

$$\boxed{A = (\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \dots, \boldsymbol{\alpha}_N)}.$$

Basis Matrix

Select m linearly independent columns of A : $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \dots, \boldsymbol{\beta}_m$.

Define $\boxed{B = (\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \dots, \boldsymbol{\beta}_m)}$. Then B is called the **basis matrix**. Since its columns are linearly independent, $\det(B) \neq 0$, so B^{-1} exists. **A basis consists of exactly m linearly independent columns of A , assuming $\text{rank}(A) = m$.**

Basic and Non-Basic Variables

The variables corresponding to $\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_m$ are called the **basic variables**: $x_{B_1}, x_{B_2}, \dots, x_{B_m}$.

The remaining $N - m$ variables are called **non-basic variables**.

Thus,

$$\boxed{\text{Number of basic variables} = m}$$

and

$$\boxed{\text{Number of non-basic variables} = N - m}.$$

For a basic solution,

$$\boxed{\text{all non-basic variables are set equal to zero.}}$$

Basic Solution and Basic Feasible Solution

Let

$$\mathbf{x}_B = \begin{bmatrix} x_{B_1} \\ x_{B_2} \\ \vdots \\ x_{B_m} \end{bmatrix}$$

be the vector of basic variables.

Since all non-basic variables are zero,

$$B\mathbf{x}_B = \mathbf{b}.$$

Therefore,

$$\boxed{\mathbf{x}_B = B^{-1}\mathbf{b}}.$$

A basic solution is feasible if

$$\mathbf{x}_B = B^{-1}\mathbf{b} \geq 0.$$

Hence,

$$\text{Basic solution} + \text{non-negativity} = \text{B.F.S.}$$

Simple procedure

$$\text{Choose basis} \rightarrow \text{set non-basic variables to zero} \rightarrow \mathbf{x}_B = B^{-1}\mathbf{b} \rightarrow \text{check } \mathbf{x}_B \geq 0.$$

Example 1. (Finding a B.F.S.) Consider

$$\begin{aligned} x_1 + x_2 + x_3 &= 4, \\ 2x_1 + x_2 + x_3 &= 5, \end{aligned} \quad x_1, x_2, x_3 \geq 0.$$

Then $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Choose the first two columns: $B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$. Since $\det(B) = -1 \neq 0$, B

is a valid basis matrix. Set the remaining variable to zero: $x_3 = 0$. Then $B \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$. Now

$B^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$. Hence $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = B^{-1} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. Therefore, $(x_1, x_2, x_3) = (1, 3, 0)$ is a B.F.S., because all variables are non-negative.

Objective Function at a B.F.S.

Let the objective coefficients of the basic variables be

$$c_{B_1}, c_{B_2}, \dots, c_{B_m}.$$

Define

$$\mathbf{c}_B = (c_{B_1}, c_{B_2}, \dots, c_{B_m}).$$

Since all non-basic variables are zero,

$$Z = c_{B_1}x_{B_1} + \dots + c_{B_m}x_{B_m}.$$

Thus, $Z = \mathbf{c}_B \mathbf{x}_B$. Since $\mathbf{x}_B = B^{-1}\mathbf{b}$, we obtain

$$Z = \mathbf{c}_B B^{-1} \mathbf{b}.$$

Expressing a Column of A in Terms of the Basis: Since $\beta_1, \dots, \beta_m$ form a basis of $\mathbb{R}^m$, every column α_j of A can be expressed as $\alpha_j = y_{1j}\beta_1 + \dots + y_{mj}\beta_m$. In matrix form, $\alpha_j = BY_j$ where

$$Y_j = \begin{bmatrix} y_{1j} \\ y_{2j} \\ \vdots \\ y_{mj} \end{bmatrix}.$$

Multiplying by B^{-1} gives $Y_j = B^{-1}\alpha_j$.

The Quantity Z_j : Define $Z_j = \mathbf{c}_B Y_j$. Therefore, $Z_j = c_{B_1} y_{1j} + \cdots + c_{B_m} y_{mj}$. Since $Y_j = B^{-1} \alpha_j$, we also have

$$Z_j = \mathbf{c}_B B^{-1} \alpha_j.$$

The quantities Z_j will be used later in the optimality test of the simplex method.

Fundamental Theorem of Linear Programming

Theorem: Consider Maximize $Z = \mathbf{c}\mathbf{x}$ subject to $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq 0$, where A is an $m \times N$ matrix with $\text{rank}(A) = m$.

If the L.P.P. has an optimal feasible solution, then at least one Basic Feasible Solution is optimal. **This means**

$$\text{Optimal feasible solution exists} \implies \text{Optimal B.F.S. exists.}$$

This is the main mathematical reason for using the simplex method.

Obtaining a B.F.S. from a Feasible Solution

Theorem: Consider Maximize $Z = \mathbf{c}\mathbf{x}$ subject to $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq 0$, where A is an $m \times (m + n)$ matrix with $\text{rank}(A) = m$.

If the L.P.P. has at least one feasible solution, then it has at least one Basic Feasible Solution.

Example 2: If $x_1 = 2$, $x_2 = 3$, $x_3 = 1$ be a feasible solution of the linear programming problem

$$\max. Z = x_1 + 2x_2 + 4x_3$$

subject to

$$\begin{aligned} 2x_1 + x_2 + 4x_3 &= 11, \\ 3x_1 + x_2 + 5x_3 &= 14, \\ x_1, x_2, x_3 &\geq 0, \end{aligned}$$

then find a basic feasible solution from the given feasible solution.

Solution: The given L.P.P. can be expressed as

$$\max. Z = x_1 + 2x_2 + 4x_3,$$

subject to

$$A\mathbf{x} = \mathbf{b}$$

or

$$\begin{bmatrix} 2 & 1 & 4 \\ 3 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 11 \\ 14 \end{bmatrix}.$$

or

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \mathbf{b}, \tag{1}$$

where

$$\alpha_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 11 \\ 14 \end{bmatrix}.$$

Since $x_1 = 2$, $x_2 = 3$, $x_3 = 1$ is a feasible solution to the given L.P.P., therefore, from (1),

$$2\alpha_1 + 3\alpha_2 + \alpha_3 = \mathbf{b}. \quad (2)$$

Now the vectors $\alpha_1, \alpha_2, \alpha_3$ associated with non-zero variables x_1, x_2, x_3 will be linearly dependent if one of these vectors can be expressed as the linear combination of the remaining two vectors.

Let

$$\alpha_1 = a\alpha_2 + b\alpha_3. \quad (3)$$

or

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} a + 4b \\ a + 5b \end{bmatrix}.$$

Thus, $a + 4b = 2$ and $a + 5b = 3 \Rightarrow a = -2, \quad b = 1.$

Substituting the values of a and b in (3), we get

$$\alpha_1 = -2\alpha_2 + \alpha_3 \text{ and } \alpha_1 + 2\alpha_2 - \alpha_3 = 0. \quad (4)$$

or $\sum_{i=1}^3 \lambda_i \alpha_i = 0$ which gives $\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = -1.$

Now we shall determine which of the three variables x_1, x_2, x_3 should be zero.

Let

$$\begin{aligned} \nu &= \max_{1 \leq i \leq 3} \left(\frac{\lambda_i}{x_i} \right) = \max \left(\frac{\lambda_1}{x_1}, \frac{\lambda_2}{x_2}, \frac{\lambda_3}{x_3} \right) \\ &= \max \left(\frac{1}{2}, \frac{2}{3}, -1 \right) = \frac{2}{3} = \frac{\lambda_2}{x_2}. \end{aligned}$$

Rule: Define

$$\nu = \max_{1 \leq i \leq n} \left(\frac{\lambda_i}{x_i} \right).$$

The variable x_i corresponding to the maximum value of $\frac{\lambda_i}{x_i}$ is set equal to zero.

Hence, x_2 should be zero, for which we should eliminate α_2 between (2) and (4).

Eliminating α_2 between (2) and (4), we get

$$2\alpha_1 - \frac{3}{2}(\alpha_1 - \alpha_3) + \alpha_2 = \mathbf{b}$$

or

$$\frac{1}{2}\alpha_1 + \frac{5}{2}\alpha_3 = \mathbf{b}.$$

Therefore, the new F.S. is

$$x_1 = \frac{1}{2}, \quad x_2 = 0, \quad x_3 = \frac{5}{2}.$$

Now the column vectors α_1 and α_3 corresponding to basic (non-zero) variables x_1 and x_3 are L.I. as

$$|(\alpha_1, \alpha_3)| = \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} = -2 \neq 0.$$

So this new F.S. is B.F.S. Hence, the B.F.S. obtained from the given F.S. is

$$\boxed{x_1 = \frac{1}{2}, \quad x_2 = 0, \quad x_3 = \frac{5}{2}}.$$

Example 3: Consider the set of equations

$$\begin{aligned}5x_1 - 4x_2 + 3x_3 + x_4 &= 3, \\2x_1 + x_2 + 5x_3 - 3x_4 &= 0, \\x_1 + 6x_2 - 4x_3 + 2x_4 &= 15,\end{aligned}$$

where

$$x_1, x_2, x_3, x_4 \geq 0.$$

If $x_1 = 1$, $x_2 = 2$, $x_3 = 1$, $x_4 = 3$ is a feasible solution, find a basic feasible solution.

Solution: The given system can be written in matrix form as

$$A\mathbf{x} = \mathbf{b},$$

where

$$\begin{bmatrix} 5 & -4 & 3 & 1 \\ 2 & 1 & 5 & -3 \\ 1 & 6 & -4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 15 \end{bmatrix}.$$

Let

$$\alpha_1 = \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} -4 \\ 1 \\ 6 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 3 \\ 5 \\ -4 \end{bmatrix}, \quad \alpha_4 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 0 \\ 15 \end{bmatrix}.$$

Then

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_4 = \mathbf{b}. \quad (1)$$

Since

$$x_1 = 1, \quad x_2 = 2, \quad x_3 = 1, \quad x_4 = 3$$

is a feasible solution, from (1) we have

$$\alpha_1 + 2\alpha_2 + \alpha_3 + 3\alpha_4 = \mathbf{b}. \quad (2)$$

Now, there are four vectors

$$\alpha_1, \alpha_2, \alpha_3, \alpha_4$$

in $\mathbb{R}^3$. Therefore, these vectors are linearly dependent.

Let

$$\alpha_1 = a\alpha_2 + b\alpha_3 + c\alpha_4. \quad (3)$$

Therefore,

$$\begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} = a \begin{bmatrix} -4 \\ 1 \\ 6 \end{bmatrix} + b \begin{bmatrix} 3 \\ 5 \\ -4 \end{bmatrix} + c \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}.$$

Comparing the corresponding components, we get

$$-4a + 3b + c = 5,$$

$$a + 5b - 3c = 2,$$

$$6a - 4b + 2c = 1.$$

Solving these equations, we obtain

$$a = \frac{22}{43}, \quad b = \frac{139}{86}, \quad c = \frac{189}{86}.$$

Therefore,

$$\frac{22}{43}\alpha_2 + \frac{139}{86}\alpha_3 + \frac{189}{86}\alpha_4 = \alpha_1.$$

Multiplying by 86, we obtain

$$86\alpha_1 - 44\alpha_2 - 139\alpha_3 - 189\alpha_4 = 0. \quad (4)$$

Thus,

$$\sum_{i=1}^4 \lambda_i \alpha_i = 0,$$

where $\lambda_1 = 86$, $\lambda_2 = -44$, $\lambda_3 = -139$, $\lambda_4 = -189$. Now define $\nu = \max_{1 \leq i \leq 4} \left(\frac{\lambda_i}{x_i} \right)$. Since

$$(x_1, x_2, x_3, x_4) = (1, 2, 1, 3),$$

we have

$$\begin{aligned} \nu &= \max \left\{ \frac{86}{1}, -\frac{44}{2}, -\frac{139}{1}, -\frac{189}{3} \right\} \\ &= \max \{86, -22, -139, -63\} \\ &= 86 = \frac{\lambda_1}{x_1}. \end{aligned}$$

Therefore, the variable x_1 should be made zero. In other words, the vector α_1 should be eliminated between (2) and (4).

From (4),

$$\alpha_1 = \frac{44}{86}\alpha_2 + \frac{139}{86}\alpha_3 + \frac{189}{86}\alpha_4.$$

Substituting this in (2), we obtain

$$\frac{44\alpha_2 + 139\alpha_3 + 189\alpha_4}{86} + 2\alpha_2 + \alpha_3 + 3\alpha_4 = \mathbf{b}.$$

Therefore,

$$0\alpha_1 + \frac{216}{86}\alpha_2 + \frac{225}{86}\alpha_3 + \frac{447}{86}\alpha_4 = \mathbf{b}.$$

Comparing this with

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_4 = \mathbf{b},$$

we obtain

$x_1 = 0, \quad x_2 = \frac{216}{86}, \quad x_3 = \frac{225}{86}, \quad x_4 = \frac{447}{86}.$
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Since

$$x_1, x_2, x_3, x_4 \geq 0,$$

the new solution is feasible.

Moreover, the three corresponding columns

$$\alpha_2, \alpha_3, \alpha_4$$

are linearly independent. Hence, the solution is a **Basic Feasible Solution**.

Therefore,

$$\left(0, \frac{216}{86}, \frac{225}{86}, \frac{447}{86}\right)$$

is a basic feasible solution.

Question 3. The point $(x_1, x_2, x_3) = (2, 4, 5)$ is a feasible solution (F.S.) of the system

$$\begin{aligned} 2x_1 - x_2 + 2x_3 &= 10, \\ x_1 + 4x_2 &= 18, \end{aligned}$$

where

$$x_1, x_2, x_3 \geq 0.$$

Reduce this F.S. to a B.F.S.

Solution

The given system can be written in matrix form as

$$A\mathbf{x} = \mathbf{b},$$

where

$$\begin{bmatrix} 2 & -1 & 2 \\ 1 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 18 \end{bmatrix}.$$

Let

$$\alpha_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 10 \\ 18 \end{bmatrix}.$$

Then

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \mathbf{b}. \tag{1}$$

Since

$$(x_1, x_2, x_3) = (2, 4, 5)$$

is a feasible solution, from (1) we have

$$2\alpha_1 + 4\alpha_2 + 5\alpha_3 = \mathbf{b}. \tag{2}$$

There are three vectors

$$\alpha_1, \alpha_2, \alpha_3$$

in $\mathbb{R}^2$. Hence, they are linearly dependent.

Let

$$\alpha_1 = a\alpha_2 + b\alpha_3. \tag{3}$$

Therefore,

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = a \begin{bmatrix} -1 \\ 4 \end{bmatrix} + b \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Comparing components, we obtain $-a + 2b = 2, 4a = 1$. Thus, $a = \frac{1}{4}$.

Substituting this into the first equation, $-\frac{1}{4} + 2b = 2$, so $2b = \frac{9}{4}$, and hence $b = \frac{9}{8}$. Therefore, $\alpha_1 = \frac{1}{4}\alpha_2 + \frac{9}{8}\alpha_3$.

Multiplying by 8, we get

$$8\alpha_1 - 2\alpha_2 - 9\alpha_3 = 0. \quad (4)$$

Thus,

$$\sum_{i=1}^3 \lambda_i \alpha_i = 0,$$

where

$$\lambda_1 = 8, \quad \lambda_2 = -2, \quad \lambda_3 = -9.$$

Now, following the method, define

$$\nu = \max_{1 \leq i \leq 3} \left(\frac{\lambda_i}{x_i} \right).$$

Since

$$(x_1, x_2, x_3) = (2, 4, 5),$$

we have

$$\begin{aligned} \nu &= \max \left\{ \frac{8}{2}, -\frac{2}{4}, -\frac{9}{5} \right\} \\ &= \max \left\{ 4, -\frac{1}{2}, -\frac{9}{5} \right\} \\ &= 4. \end{aligned}$$

Hence,

$$\nu = \frac{\lambda_1}{x_1}.$$

Therefore, the variable corresponding to λ_1 , namely x_1 , should be made zero.

Thus,

$$\boxed{x_1 = 0}.$$

From (4),

$$8\alpha_1 = 2\alpha_2 + 9\alpha_3,$$

and hence

$$\alpha_1 = \frac{1}{4}\alpha_2 + \frac{9}{8}\alpha_3.$$

Substituting this in (2), we get

$$2 \left(\frac{1}{4}\alpha_2 + \frac{9}{8}\alpha_3 \right) + 4\alpha_2 + 5\alpha_3 = \mathbf{b}.$$

Therefore,

$$\left(\frac{1}{2} + 4 \right) \alpha_2 + \left(\frac{9}{4} + 5 \right) \alpha_3 = \mathbf{b}.$$

Hence,

$$\frac{9}{2}\alpha_2 + \frac{29}{4}\alpha_3 = \mathbf{b}.$$

Comparing this with

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \mathbf{b},$$

we obtain

$$\boxed{x_1 = 0, \quad x_2 = \frac{9}{2}, \quad x_3 = \frac{29}{4}.}$$

Therefore, the new solution is

$$\boxed{\left(0, \frac{9}{2}, \frac{29}{4}\right)}.$$

Since

$$x_1 = 0, \quad x_2 = \frac{9}{2} > 0, \quad x_3 = \frac{29}{4} > 0,$$

the solution is feasible.

Moreover, the corresponding columns

$$\alpha_2, \alpha_3$$

are linearly independent. Hence, the new feasible solution is a **Basic Feasible Solution**.

Therefore,

$$\boxed{\left(0, \frac{9}{2}, \frac{29}{4}\right) \text{ is a B.F.S.}}$$

Example 4. The point

$$(1, 1, 1)$$

is a feasible solution (F.S.) of the system

$$\begin{aligned} x_1 + x_2 + 2x_3 &= 4, \\ 2x_1 - x_2 + x_3 &= 2, \end{aligned}$$

where

$$x_1, x_2, x_3 \geq 0.$$

Reduce the F.S. to a B.F.S.

Solution

The given system can be written in matrix form as

$$A\mathbf{x} = \mathbf{b},$$

where

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

Let

$$\alpha_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

Then

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \mathbf{b}. \tag{5}$$

Since

$$(x_1, x_2, x_3) = (1, 1, 1)$$

is a feasible solution, from (5),

$$\alpha_1 + \alpha_2 + \alpha_3 = \mathbf{b}. \quad (6)$$

There are three vectors

$$\alpha_1, \alpha_2, \alpha_3$$

in $\mathbb{R}^2$. Hence, they are linearly dependent.

Observe that

$$\alpha_1 + \alpha_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \alpha_3.$$

Therefore,

$$\alpha_1 + \alpha_2 - \alpha_3 = 0. \quad (7)$$

Thus,

$$\sum_{i=1}^3 \lambda_i \alpha_i = 0,$$

where

$$\lambda_1 = 1, \quad \lambda_2 = 1, \quad \lambda_3 = -1.$$

Now define

$$\nu = \max_{1 \leq i \leq 3} \left(\frac{\lambda_i}{x_i} \right).$$

Since

$$(x_1, x_2, x_3) = (1, 1, 1),$$

we have

$$\begin{aligned} \nu &= \max \left\{ \frac{1}{1}, \frac{1}{1}, -\frac{1}{1} \right\} \\ &= 1. \end{aligned}$$

Here,

$$\frac{\lambda_1}{x_1} = \frac{\lambda_2}{x_2} = 1.$$

Thus, there is a **tie**: either x_1 or x_2 may be chosen to leave the basis.

Following the method, choose

$$\boxed{x_1 = 0}.$$

From (7),

$$\alpha_1 = -\alpha_2 + \alpha_3.$$

Substituting this into (6), we obtain

$$(-\alpha_2 + \alpha_3) + \alpha_2 + \alpha_3 = \mathbf{b}.$$

Hence,

$$2\alpha_3 = \mathbf{b}.$$

Comparing this with

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \mathbf{b},$$

we obtain

$$\boxed{x_1 = 0, \quad x_2 = 0, \quad x_3 = 2.}$$

Therefore,

$$(x_1, x_2, x_3) = (0, 0, 2).$$

This is a feasible solution. The basic column is

$$\alpha_3,$$

and it is nonzero, so it forms a basis of $\mathbb{R}^2$ only after accounting for the fact that the standard basis definition requires $m = 2$ basic columns.

Thus, $(0, 0, 2)$ is a **degenerate B.F.S.** when represented with two basic variables, one of which has value zero.

Alternatively, because the ratio test has a tie, we may choose the opposite direction of the dependence relation:

$$-\alpha_1 - \alpha_2 + \alpha_3 = 0.$$

Using the corresponding direction gives the other B.F.S.

$$(x_1, x_2, x_3) = (2, 2, 0).$$

Indeed,

$$2 + 2 + 2(0) = 4,$$

and

$$2(2) - 2 + 0 = 2.$$

Hence,

$$(2, 2, 0) \text{ is also a B.F.S.}$$

Therefore, a convenient B.F.S. is

$$(2, 2, 0).$$

Example 5: The point

$$x_1 = 2, \quad x_2 = 4, \quad x_3 = 5$$

is a feasible solution (F.S.) of the system of equations

$$2x_1 - x_2 + 2x_3 = 10,$$

$$x_1 + 4x_2 = 18,$$

where

$$x_1, x_2, x_3 \geq 0.$$

Reduce this F.S. to a B.F.S.

Solution: The given system can be written in matrix form as

$$A\mathbf{x} = \mathbf{b},$$

where

$$\begin{bmatrix} 2 & -1 & 2 \\ 1 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 18 \end{bmatrix}.$$

Let

$$\boldsymbol{\alpha}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \boldsymbol{\alpha}_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \quad \boldsymbol{\alpha}_3 = \begin{bmatrix} 2 \\ 0 \end{bmatrix},$$

and

$$\mathbf{b} = \begin{bmatrix} 10 \\ 18 \end{bmatrix}.$$

Therefore,

$$\boldsymbol{\alpha}_1 x_1 + \boldsymbol{\alpha}_2 x_2 + \boldsymbol{\alpha}_3 x_3 = \mathbf{b}. \quad (1)$$

Since

$$x_1 = 2, \quad x_2 = 4, \quad x_3 = 5$$

is a feasible solution, from (1) we have

$$2\boldsymbol{\alpha}_1 + 4\boldsymbol{\alpha}_2 + 5\boldsymbol{\alpha}_3 = \mathbf{b}. \quad (2)$$

Now, there are three vectors

$$\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \boldsymbol{\alpha}_3$$

in $\mathbb{R}^2$. Hence, they are linearly dependent.

Let

$$\boldsymbol{\alpha}_1 = a\boldsymbol{\alpha}_2 + b\boldsymbol{\alpha}_3. \quad (3)$$

Thus,

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = a \begin{bmatrix} -1 \\ 4 \end{bmatrix} + b \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Comparing the corresponding components, we get

$$-a + 2b = 2, \quad (4)$$

and

$$4a = 1. \quad (5)$$

From (5),

$$a = \frac{1}{4}.$$

Substituting this in (4),

$$-\frac{1}{4} + 2b = 2.$$

Therefore,

$$2b = \frac{9}{4},$$

and hence

$$b = \frac{9}{8}.$$

Thus,

$$\boldsymbol{\alpha}_1 = \frac{1}{4}\boldsymbol{\alpha}_2 + \frac{9}{8}\boldsymbol{\alpha}_3.$$

Multiplying by 8,

$$8\boldsymbol{\alpha}_1 - 2\boldsymbol{\alpha}_2 - 9\boldsymbol{\alpha}_3 = 0.$$

Multiplying by -1 , we write

$$-8\alpha_1 + 2\alpha_2 + 9\alpha_3 = 0. \quad (6)$$

Thus,

$$\sum_{i=1}^3 \lambda_i \alpha_i = 0,$$

where

$$\lambda_1 = -8, \quad \lambda_2 = 2, \quad \lambda_3 = 9.$$

Now, by the rule,

$$\nu = \max_{1 \leq i \leq 3} \left(\frac{\lambda_i}{x_i} \right).$$

Therefore,

$$\begin{aligned} \nu &= \max \left\{ \frac{-8}{2}, \frac{2}{4}, \frac{9}{5} \right\} \\ &= \max \left\{ -4, \frac{1}{2}, \frac{9}{5} \right\} \\ &= \frac{9}{5}. \end{aligned}$$

Thus,

$$\nu = \frac{\lambda_3}{x_3}.$$

Hence, the variable x_3 is set equal to zero:

$$\boxed{x_3 = 0}.$$

From (6),

$$-8\alpha_1 + 2\alpha_2 + 9\alpha_3 = 0.$$

Therefore,

$$9\alpha_3 = 8\alpha_1 - 2\alpha_2,$$

and hence

$$\alpha_3 = \frac{8}{9}\alpha_1 - \frac{2}{9}\alpha_2. \quad (7)$$

We now eliminate α_3 between (2) and (7).

Substituting (7) into (2), we get

$$2\alpha_1 + 4\alpha_2 + 5 \left(\frac{8}{9}\alpha_1 - \frac{2}{9}\alpha_2 \right) = \mathbf{b}.$$

Therefore,

$$2\alpha_1 + 4\alpha_2 + \frac{40}{9}\alpha_1 - \frac{10}{9}\alpha_2 = \mathbf{b}.$$

Collecting the coefficients of α_1 and α_2 ,

$$\left(2 + \frac{40}{9} \right) \alpha_1 + \left(4 - \frac{10}{9} \right) \alpha_2 = \mathbf{b}.$$

Hence,

$$\frac{58}{9}\alpha_1 + \frac{26}{9}\alpha_2 = \mathbf{b}. \quad (8)$$

Comparing (8) with

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \mathbf{b},$$

we obtain

$$x_1 = \frac{58}{9}, \quad x_2 = \frac{26}{9}, \quad x_3 = 0.$$

Since

$$\frac{58}{9} > 0, \quad \frac{26}{9} > 0, \quad x_3 = 0,$$

the solution is feasible.

The two corresponding columns

$$\alpha_1, \alpha_2$$

are linearly independent. Therefore, the above feasible solution is a basic feasible solution.

Hence,

$$\left(\frac{58}{9}, \frac{26}{9}, 0 \right) \text{ is a B.F.S.}$$

Example 6: The point

$$(x_1, x_2, x_3) = (1, 1, 1)$$

is a feasible solution (F.S.) of the system

$$\begin{aligned} x_1 + x_2 + 2x_3 &= 4, \\ 2x_1 - x_2 + x_3 &= 2, \end{aligned}$$

where

$$x_1, x_2, x_3 \geq 0.$$

Reduce this F.S. to a B.F.S.

Solution: The given system can be written in matrix form as

$$A\mathbf{x} = \mathbf{b},$$

where

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

Let

$$\alpha_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

and

$$\mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

Therefore,

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \mathbf{b}. \tag{9}$$

Since

$$(x_1, x_2, x_3) = (1, 1, 1)$$

is a feasible solution, from (9) we have

$$\alpha_1 + \alpha_2 + \alpha_3 = \mathbf{b}. \tag{10}$$

There are three vectors

$$\alpha_1, \alpha_2, \alpha_3$$

in $\mathbb{R}^2$. Hence, they are linearly dependent.

Observe that

$$\alpha_1 + \alpha_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \alpha_3.$$

Therefore,

$$\alpha_1 + \alpha_2 - \alpha_3 = 0. \quad (11)$$

Thus,

$$\sum_{i=1}^3 \lambda_i \alpha_i = 0,$$

where

$$\lambda_1 = 1, \quad \lambda_2 = 1, \quad \lambda_3 = -1.$$

Using the rule,

$$\nu = \max_{1 \leq i \leq 3} \left(\frac{\lambda_i}{x_i} \right).$$

Since

$$x_1 = x_2 = x_3 = 1,$$

we have

$$\begin{aligned} \nu &= \max \left\{ \frac{1}{1}, \frac{1}{1}, -\frac{1}{1} \right\} \\ &= \max \{1, 1, -1\} \\ &= 1. \end{aligned}$$

Here the maximum value occurs twice:

$$\frac{\lambda_1}{x_1} = \frac{\lambda_2}{x_2} = 1.$$

Therefore, either x_1 or x_2 can be selected to become zero.

First B.F.S.: Choose

$$\boxed{x_1 = 0}.$$

From (11),

$$\alpha_1 = -\alpha_2 + \alpha_3. \quad (12)$$

Substituting (12) into (10), we obtain

$$(-\alpha_2 + \alpha_3) + \alpha_2 + \alpha_3 = \mathbf{b}.$$

Therefore,

$$2\alpha_3 = \mathbf{b}.$$

Comparing with

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \mathbf{b},$$

we obtain

$$\boxed{x_1 = 0, \quad x_2 = 0, \quad x_3 = 2.}$$

Thus,

$$\boxed{(0, 0, 2)}$$

is a basic feasible solution.

Second B.F.S.: Since the maximum value

$$\nu = 1$$

is attained for both $i = 1$ and $i = 2$, we may alternatively choose

$$\boxed{x_2 = 0}.$$

From (11),

$$\alpha_2 = -\alpha_1 + \alpha_3. \quad (13)$$

Substituting (13) into (10), we get

$$\alpha_1 + (-\alpha_1 + \alpha_3) + \alpha_3 = \mathbf{b}.$$

Hence,

$$2\alpha_3 = \mathbf{b}.$$

Thus, again,

$$x_3 = 2.$$

However, this gives

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 2,$$

which is the same B.F.S. as above.

To obtain the other B.F.S. listed in the question, we use the other possible direction of the same dependence relation:

$$-\alpha_1 - \alpha_2 + \alpha_3 = 0. \quad (14)$$

Thus,

$$\lambda_1 = -1, \quad \lambda_2 = -1, \quad \lambda_3 = 1.$$

Then

$$\nu = \max \{-1, -1, 1\} = 1 = \frac{\lambda_3}{x_3}.$$

Hence,

$$x_3 = 0.$$

From (14),

$$\alpha_3 = \alpha_1 + \alpha_2.$$

Substituting this in (10),

$$\alpha_1 + \alpha_2 + (\alpha_1 + \alpha_2) = \mathbf{b}.$$

Therefore,

$$2\alpha_1 + 2\alpha_2 = \mathbf{b}.$$

Comparing with

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \mathbf{b},$$

we obtain

$$\boxed{x_1 = 2, \quad x_2 = 2, \quad x_3 = 0.}$$

Therefore,

$$(2, 2, 0)$$

is another basic feasible solution.

Hence, the two B.F.S.s are

$$(0, 0, 2) \quad \text{and} \quad (2, 2, 0).$$